

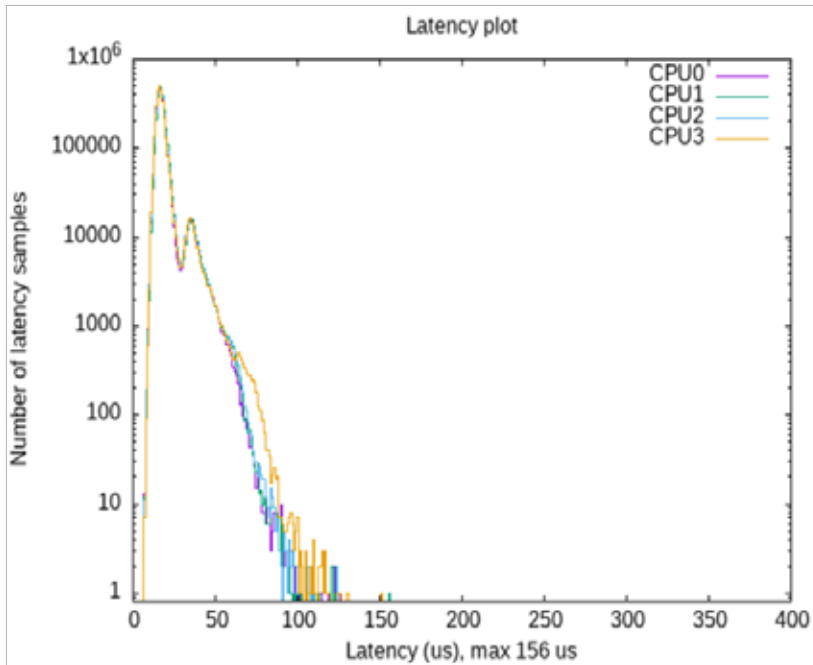


Figure 9: Graph of Preemptible Kernel ((real-time))

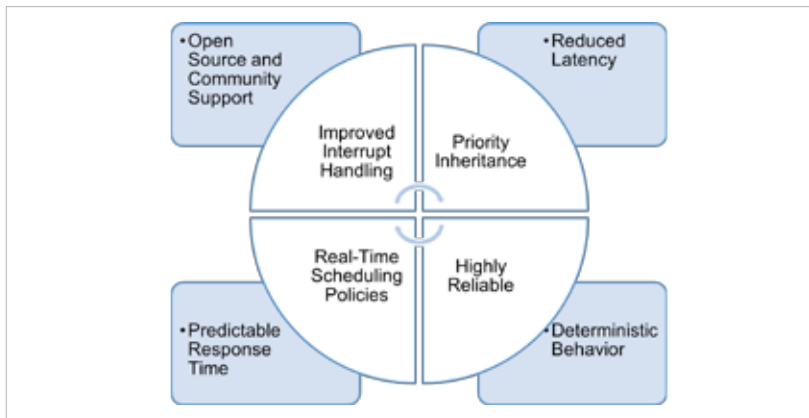


Figure 10: PREEMPT_RT patch benefits

- **Error handling:** Implement robust error handling and fault tolerance mechanisms in your real-time application to handle unexpected situations and recover gracefully.

Comparison of QNX and Real-Time Linux

QNX and Real-Time Linux are both popular choices for building real-time and embedded systems, but they have different characteristics, features, and performance considerations. Table 3 gives a comparison of QNX and Real-Time Linux in terms of real-time performance.

In making your decision, consider the following factors.

Timing requirements: PREEMPT-RT patch converts standard Linux to Real-Time Linux. The latter has good determinism to achieve real-time capabilities and reduces scheduling latency. However, QNX may be slightly better compared to Real-Time Linux if your application requires hard real-time capabilities and precise determinism, though the latter can be used based on the requirements.

Reliability and safety: QNX is often used in safety-critical systems including medical devices. It is designed with safety in mind and provides features for safety certification. Real-Time Linux can be used in safety-critical applications, but achieving safety certification may require additional

Characteristics	QNX	Real-Time Linux
Determinism	Better determinism and real-time performance out of the box due to its microkernel architecture	Real-Time Linux can achieve good determinism with proper configuration and patches
Latency	Has lower latencies and faster context switching, making it suitable for hard real-time applications	Achieves low latencies with specialised patches but might have slightly higher latencies compared to QNX
Ease of use	QNX is designed specifically for real-time applications and provides a comprehensive environment for embedded systems development	Real-Time Linux may require more configuration and tuning to achieve optimal real-time performance
Ecosystem and compatibility	QNX has a more focused ecosystem tailored to embedded and real-time applications.	Real-Time Linux benefits from the vast Linux ecosystem, including a wide range of device drivers and software libraries

Table 3: Comparison of QNX and Real-Time Linux